

Chapter

Body Types

1. Which body type has a high metabolic rate? MAIN 2024

- (a) Mesomorph
- (b) Endomorph
- (c) Ectomorph
- (d) Obese

2. Love of food and comfort is a psychological characteristic of which body type?

- (a) Endomorph
- (b) Mesomorph
- (c) Ectomorph
- (d) Hard Gainer

3. Ectomorph body usually refers to: SQP 2023

- (a) Rounded physique
- (b) Balanced body
- (c) Lean body
- (d) Muscular body

4. Which body type has a rounded physique, higher fat percentage, and shorter limbs? SQP 2022

- (a) Mesomorph
- (b) Ectomorph
- (c) Endomorph
- (d) None of the above

5. What is a defining characteristic of endomorphs? MAIN 2022

- (a) Low body fat percentage
- (b) Large bone structure and high-fat retention
- (c) Easily gain muscle with low effort
- (d) Fast metabolism

6. Which macronutrient should endomorphs prioritize? SQP 2023

- (a) High carbohydrates
- (b) High sugar intake
- (c) Protein and fiber
- (d) No protein intake

**7. Which body type is naturally lean with weak muscle development?
MAIN 2021**

- (a) Mesomorph
- (b) Ectomorph
- (c) Endomorph
- (d) None of the above

8. What is the primary goal for endomorphs in fitness training? SQP 2024

- (a) Muscle gain
- (b) Fat loss and weight management
- (c) Avoiding physical activity
- (d) Gaining weight quickly

**9. ____ is the most effective exercise method for endomorphs to lose weight.
SQP 2023**

- (a) Sprinting only
- (b) Low-intensity cardio combined with strength training
- (c) Only body weight exercises
- (d) Avoiding exercise and focusing on diet alone

10. ____ is the main reason why mesomorphs are considered the most athletic body type. MAIN 2021

- (a) Naturally high fat percentage
- (b) Balanced metabolism and muscular structure
- (c) Weak muscle development
- (d) Difficulty in gaining weight

11. Assertion (A) : Endomorphs tend to gain weight easily. Reason (R) : Endomorphs have a slow metabolism and store fat quickly. COMP 2024

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

12. Assertion (A) : Mesomorphs should follow a diet that includes equal amounts of protein and fat, with moderate carbohydrates.

Reason (R) : Mesomorphs require a balanced diet to support their naturally muscular build. SQP 2022

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

13. Assertion (A) : Mesomorphs can easily gain or lose weight.

Reason (R) : Mesomorphs have a well-balanced metabolism and muscular body structure. MAIN 2023

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

14. Match the following training goals with the recommended exercise types:

Column I (Training Goal)		Column II (Exercise Type)	
(I)	Fat loss	1.	Strength training
(II)	Muscle gain	2.	High-Intensity Interval Training (HIIT)
(III)	Endurance improvement	3.	Long-distance running

COMP 2021

- (a) I-1, II-2, III-3
- (b) I-2, II-1, III-3
- (c) I-3, II-1, III-2
- (d) I-2, II-3, III-1

15. Match the following training programs with their recommended body types:

Column I (Training Program)		Column II (Best for)	
(I)	HIIT	1.	Endomorph
(II)	Heavy Strength Training	2.	Ectomorph
(III)	Mixed Strength & Cardio	3.	Mesomorph

MAIN 2023

- (a) I-1, II-2, III-3
- (b) I-2, II-3, III-1
- (c) I-3, II-1, III-2
- (d) I-1, II-3, III-2

16. Match the following training techniques with their benefits:

Column I (Training Technique)		Column II (Benefit)	
(I)	Compound Movements	1.	Engages multiple muscle groups for strength gain
(II)	Isolation Exercises	2.	Targets specific muscles for definition
(III)	Functional Training	3.	Improves overall mobility and coordination

- (a) I-1, II-2, III-3
- (b) I-2, II-3, III-1
- (c) I-3, II-1, III-2
- (d) I-1, II-3, III-2

SQP 2022

17. Rahul has a naturally broad and muscular frame. He finds it easy to gain both muscle and fat. His coach advises him to maintain a balanced workout routine that includes both weight training and aerobic exercises.

(i) Which body type does Rahul belong to?

(ii) What type of training should he follow to maintain overall fitness?

MAIN 2024

18. A sports coach is conducting a session on body types. He explains that one group of people can store fat easily and require a stricter eating plan, while another group has a fast metabolism and struggles to gain weight.

(i) Which two body types are being discussed?

(ii) How do their dietary needs differ?

SQP 2021

19. Simran, a fitness enthusiast, follows a highprotein diet and performs heavy compound lifts like deadlifts and squats to gain muscle mass. She avoids excessive cardio to prevent unnecessary calorie burning.

(i) Which body type does Simran belong to?

(ii) Why should she limit excessive cardio exercises?

COMP 2023

20. Kunal has a slow metabolism and struggles to lose fat. His nutritionist advises him to avoid simple carbohydrates like white bread and focus on consuming more proteins and complex carbs.

(i) Which body type does Kunal have?

(ii) Why is it important for him to avoid simple carbohydrates?

SQP 2022

21. Explain the classification of individual body types as given by William Herbert Sheldon.

SQP 2022

22. What are the traits of Ectomorphs?

MAIN 2024

23. State the characteristic features of Endomorphs.

SQP 2021

24. How can an ectomorph effectively gain muscle?

SQP 2022

25. What are the differences in the training requirements of Mesomorphs and Ectomorphs? SQP 2021

26. How do the nutritional requirements of Endomorphs differ from those of Ectomorphs? COMP 2023

Solution

1. (c) Ectomorph

Ectomorphs are generally **thyroid-dominant**, giving them a **higher basal metabolic rate (BMR)**. BMR is the energy the body burns at rest, making ectomorphs naturally leaner and faster in metabolism.

2. (b) Mesomorph

Mesomorphs are characterized by broad shoulders, large heads, narrow waists, and a muscular chest, arms, and legs. They value food and comfort.

3. (c) Lean body

Ectomorphs are **thin, lanky, and lean**. They often have **long limbs, small muscles**, and find it difficult to gain weight even with a high-calorie diet.

4. (c) Endomorph

Explanation: Endomorphs have a **soft, rounded body, shorter arms and legs**, and a **higher body fat percentage**. They store fat easily and may struggle with weight management without proper diet and exercise.

5. (b) Large bone structure and high-fat retention

Endomorphs have a **broader bone structure** and tend to **store fat easily**. Without proper diet and exercise, they are prone to obesity. However, with **resistance training and metabolic workouts**, they can build strength effectively.

6. (c) Protein and fiber

Endomorphs should focus on **high-protein foods** to support **muscle retention and metabolism**, and increase **fiber intake** to promote **satiety and digestive health**. Reducing simple carbohydrates and unhealthy fats helps maintain **optimal weight** and energy levels.

7. (b) Ectomorph

Explanation: Ectomorphs are **naturally thin and lean**, with **long limbs and small muscles**. They often struggle to gain weight or build muscle despite consuming a high-calorie diet.

8. (b) Fat loss and weight management

Endomorphs **store fat easily**, so the main focus is **weight management**. Combining **strength training and cardio** (HIIT or LISS) helps **burn fat while maintaining lean muscle**, and a diet **low in carbs and high in protein** prevents excessive fat gain.

9. (b) Low-intensity cardio combined with strength training

Endomorphs benefit most from combining **LISS (low-intensity steady-state) cardio** with **strength training**. Strength exercises boost **metabolism** and preserve muscle, while LISS and HIIT support **fat loss**. A **consistent workout routine** helps prevent excessive fat accumulation.

10. (b) Balanced metabolism and muscular structure

Mesomorphs have a **naturally muscular and athletic build** with a **balanced metabolism**, making it easier for them to gain muscle, maintain low body fat, and perform well in physical activities.

11. (a) Both A and R are true, and R is the correct explanation of A.

Endomorphs have a **higher body fat percentage** and a **slow metabolism**, which makes them prone to **weight gain**. Their bodies are designed for **fat storage**, so to maintain a lean physique, they need **fat-burning exercises** like HIIT and a **low-carb, high-protein diet**. Without proper exercise and diet, they store excess calories as fat more easily than mesomorphs or ectomorphs.

12. (a) Both A and R are true, and R is the correct explanation of A.

Mesomorphs naturally build **muscle easily**, but maintaining a healthy **muscle-to-fat ratio** requires a **balanced diet**. A diet with **equal protein and fat**, and **moderate carbohydrates** provides nutrients for **muscle recovery**, **hormone regulation**, and **sustained energy**, preventing fat accumulation. Unlike

ectomorphs, they do not need excessive calories, and unlike endomorphs, they do not need to severely restrict carbohydrates.

13. (a) Both A and R are true, and R is the correct explanation of A.

Mesomorphs have a **naturally athletic build** and a **balanced metabolism**, allowing them to **gain or lose weight easily**. This makes it easier to adapt to different training programs, whether for muscle building or fat loss. Unlike endomorphs, they do not gain fat easily, and unlike ectomorphs, they can build muscle without much difficulty. Proper diet and training help maintain their muscular structure and optimal body composition.

14. (b) I-2, II-1, III-3

Fat loss is best achieved with HIIT, which burns calories effectively (I-2).

Muscle gain requires strength training to build and maintain muscle mass (II-1).

Endurance improvement is achieved with longdistance running, which enhances stamina (III-3).

15. (a) I-1, II-2, III-3

HIIT is ideal for endomorphs to burn fat efficiently (I-1).

Heavy strength training helps ectomorphs build muscle mass (II-2).

Mesomorphs benefit from a balanced mix of strength training and cardio (III-3).

16. (a) I-1, II-2, III-3

Compound movements work multiple muscles, improving strength (I-1).

Isolation exercises target specific muscles, enhancing definition (II-2).

Functional training improves mobility and coordination, benefiting overall fitness (III-3).

17. (i) Mesomorph – Rahul has a naturally athletic and muscular physique, which allows him to gain or lose weight easily.

(ii) Balanced training – He should follow a combination of **strength training** (to maintain and build muscle) and **aerobic/endurance exercises** (to control fat and improve cardiovascular fitness) to maintain overall fitness.

18. (i) **Endomorphs** (store fat easily) and **Ectomorphs** (struggle to gain weight).

(ii) **Dietary needs:**

- **Endomorphs:** Low-carbohydrate, high-protein diet to control fat accumulation and maintain lean muscle.
- **Ectomorphs:** Higher-calorie diet with complex carbohydrates and protein to support muscle gain and energy needs.

19. (i) **Ectomorph** – She has a naturally lean structure and finds it difficult to gain muscle.

(ii) **Reason to limit cardio:** Ectomorphs have a fast metabolism, so excessive cardio can burn additional calories, making it harder to gain and maintain muscle mass.

20. (i) **Endomorph** : He has a higher fat storage capacity and a slower metabolism.

(ii) Simple carbohydrates cause quick spikes in blood sugar, leading to fat accumulation, which is not ideal for endomorphs who already store fat easily.

21. William Herbert Sheldon classified human body types into three categories based on physical characteristics:

1. **Ectomorphs** – Slim and lean individuals with a fast metabolism. They have long limbs, small muscles, and a low body-fat percentage. Ectomorphs are generally suited for endurance activities but find it difficult to gain weight or muscle.
2. **Mesomorphs** – Naturally muscular and well-built individuals with a balanced metabolism. They gain muscle and strength easily and are considered the most athletic body type, suitable for activities requiring physical strength and agility.
3. **Endomorphs** – Individuals with a rounded physique, broader bone structure, and higher fat storage. They gain weight easily and require careful diet and exercise to maintain a healthy body composition.

22. Traits of Ectomorphs:

- Slim and lean physique with elongated limbs.
- Weak body constitution and low muscle mass.
- Flat chest and small bone structure.
- Struggle to gain weight and build muscle.
- Naturally excel in endurance sports due to their light frame.

23. Characteristic Features of Endomorphs:

- Rounded physique with short arms and legs.
- High-fat storage, especially in the upper body.
- Underdeveloped muscles.
- Tend to gain weight easily and struggle to lose it.
- Slower metabolism compared to other body types.

24. How an Ectomorph Can Effectively Gain Muscle:

1. **High-Calorie Diet:** Consume a calorie surplus including protein, complex carbohydrates, and healthy fats to support muscle growth.
2. **Strength Training Focus:** Prioritize heavy compound exercises such as squats, deadlifts, and bench presses to stimulate multiple muscle groups.
3. **Minimal Cardio:** Limit excessive cardio since it burns calories that are needed for muscle gain.

25. Differences in Training Requirements of Mesomorphs and Ectomorphs:

(i) Mesomorphs:

- Benefit from a combination of strength training and aerobic exercises.
- Focus on balanced workouts to maintain their naturally muscular build and optimal weight.

(ii) Ectomorphs:

- Focus primarily on compound exercises to build muscle mass.
- Avoid isolated movements; prioritize exercises targeting large muscle groups, such as bench press, deadlifts, and squats.

26. Nutritional Requirements of Endomorphs vs. Ectomorphs:

Ectomorphs:

- Can consume more carbohydrates compared to other body types.
- Should prioritize complex carbs like brown rice, whole grains, and oats for sustained energy.
- Require high protein intake to support muscle growth due to their naturally lean structure and fast metabolism.

Endomorphs:

- Require fewer carbohydrates and higher protein intake.
- Should avoid simple carbs like white bread, sugar, and pastries; focus on complex carbs for better energy management.
- Need a stricter diet to prevent excessive weight gain due to their slow metabolism and higher fat storage tendency.